FACULTY OF ENGINEERING AND ARCHITECTURE

DEPARTMENT ELECTRONICS AND INFORMATION SYSTEMS
RESEARCH GROUP IDLAB - AIDA

Prof. dr. Jefrey Lijffijt
Professor of Data Science, Knowledge Discovery, and Visual Analytics

E jefrey.lijffijt@ugent.be
T +32 4 7453 9464

www.ugent.be

Jefrey Lijffijt
Ghent University - IDLab
Technologiepark Zwijnaarde 122
9052 Ghent
Belgium

Reading report doctoral thesis Maxime Deforche

Dear Chair, dear Maxime, dear promotors, dear members of the Examination Committee,

Please find in this letter my valuation of Maxime Deforche's PhD Thesis "Management and Analysis of Short Texts: A Graph-Based Approach".

Brief Description of Contents. The research focuses on several research questions around how to efficiently and functionally represent and analyse short text data and specifically the Database of Byzantine Book Epigrams (DBBE). The research is laid out over four research questions, namely (paraphrased):

- Can graph databases be used to represent short texts efficiently and suitably?
- How to measure orthographic similarity effectively at different granularities?
- How to integrate metadata into the similarity while accounting for veracity problems?
- How can the representation and similarities be used to support
    - Analysis of patterns and connections present in the data
    - Improving the informativeness and transparency of query results
    - Clustering with the metadata under a choice on required reliability.

The thesis consists of an introduction, conclusion, and five chapters in which the research questions are answered, based fully on papers published in several journals and conference proceedings. Notably, one of the papers has received a best paper award.

Significance and Potential Impact. The work is part of a larger project called 'Interconnected Texts' and substantially contributes to several parts of this project. The advantages and capabilities gained through the contributions on the DBBE data specifically are clearly described in the thesis.

It is claimed that the contributions are also generically applicable to other short text data. I do agree that this can be expected, but I also find it a real pity that other data has not been tested in any of the contributions. Also, it is not so clear to me why only a subset of the epigrams is

used throughout. Scalability is discussed as a challenge, but I don't recall reading an explicit comment that this was the reason and probably it isn't because it seems feasible to handle a much larger set of epigrams without any problem, or wouldn't it?

Presentation. The presentation is generally of very high quality, both accessible to anyone with a relevant background, but also precise. The text shows that the candidate has an excellent command of English and academic writing, and figures and tables are used appropriately for illustrations and tabulation of information. The thesis is generally a pleasure to read and the structure is well-chosen, with a clear buildup: from the graph representation, to similarity on the graph, then similarity including metadata, and then two analysis tasks (ranking and clustering).

Questions and remarks.

Suggestions for improvement:

1. The summary gives an overview that is useful if one already understand the topics discussed in the thesis, but for me it was not very accessible. Depending on your intent with this section, the language could be simplified. The third paragraph is a bit dense with complicated sentences that an AI would write. This is in contrast with rest of the thesis which is presented very well and in an accessible manner.
    a. In the 'Samenvatting', 'volume' should probably be italic like the others.
2. Section 1 is a bit slow to get to the point of the specific challenges on short text and the DBBE data. Up to Section 1.2.2. the text remains quite generic and I did not really understand where we were heading yet.

Potential questions for during the defense:

1. In Section 1.2.2 you write that larger, more complex models bring a tension between predictive performance and interpretability. Is this really always so? Wouldn't ML models that are too simple to model the data well also be less interpretable?
2. You state as an advantage that the methods do not require training. However, non-parametric modelling as used here has two main drawbacks: there cannot be sub-linear scaling in computational complexity and these models are in fact quite large, having a number of parameters linear in the size of the data. Do you think that parametric models could provide more scalable and interpretable alternatives?
3. In 1.2.3 you seem to hint at that foundation models do not really generalize over languages and that they are not good yet for less spoken languages. Is this really so, or is it more like particularly for old greek this is the case?
4. In 1.3, you describe some of the challenges of the epigrams. Are there challenges and potential patterns that may be highly relevant and interesting that cannot be captured and identified by the proposed approach (LPG, similarities, etc.)?
5. Several parts of the thesis: shouldn't similarity always be defined with respect to a purpose?

6. In several chapters you compare against hand-annotated labels. If the purpose is to find a computational procedure that matches these as well as possible, is a hand-crafted similarity the best approach to achieve that?
7. The 'ground truth' is not recovered perfectly. Would you be willing to defend that the similarity measures that you have introduced can also bring new information beyond the variant/type annotation? If you would want to test this more formally, how would you approach that?
8. In 2.3.1, the character sequence comprising a text is indexed *1* to *n*, while the substring representation is indexed *0* to *m-1*. Is this a deliberate choice, or just an unintential inconsistency?
9. There are visualizations of graphs throughout the thesis. I could not immediately find a description of how the layout for the visualizations was determined. Mostly by hand? In Figure 3.9, Neo4j Bloom is used, do you know what algorithm is used for the layout and for the visualization to be useful, do you think the algorithm could matter?
10. You write that the performance estimates are 'best case' numbers, as the parameters/weights are tuned (e.g., in Chapter 4 you employ grid search) and you report the best cases. But isn't it then always overly optimistic rather than optimistic, as you developed these methods also to be applicable to other data? And also here, would you argue that in this setting the weights are indeed the 'best' for this data and purpose? If the intent is to let experts tune this for their setting, how do you recommend they do so?

I look forward to the internal defense to discuss the thesis and contributions with Maxime.

Best wishes,



Jefrey Lijffijt